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[Paper 1] Flagship Paper (published) - 1e11q DM Run

In the Flagship paper, Working Group I intends to include a panel of snapshots of the fiducial runs (isolated + cosmological) performed by each of the participating codes. In particular, we need a volunteer from each of the participating codes who can run a quick dark matter only cosmological simulation (described below) and generate a few images and profiles.

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[NOTE] For the latest result, please see the draft of the AGORA Flagship paper posted in [this page](#).

[NOTE] This page is for the cosmological run only. Please suggest your change in the text with a distinctive color. You may also post your comments **at the bottom of this page**.

1. Goals of The "Flagship" Run

- (1) This is a proof-of-concept simulation in a cosmological, dark matter only setup to demonstrate the capability of the collaboration.
- (2) Desired outcome: snapshots of dark matter distribution at low redshift across all participating codes

2. Overview of the Initial Condition

- (1) Dark matter only simulation with a $\sim 10^{11} M_{\text{Sun}}$ halo with a quiescent merger history
- (2) MUSIC parameter file: [this link](#) (WG III's Workspace)
 - new as of Nov. 20, 2012
 - currently defaults to HRC-3 (see below)
 - the IC used in Paper "1" has been superseded by a even newer parameter set in Paper "2" as of Jan. 17, 2014
- (3) Change levelmax=12 to 13 to increase particle mass resolution
- (4) Try changing align_top = yes if not using Enzo
- (5) Target halo mass measured in a test run at $z=0$ (4096^3 effective resolution, coarse refinement):

$$M_{\text{vir}} \sim 1.8 \times 10^{11} M_{\text{Sun}} \text{ (see note 1), or } M_{200} \sim 1.7 \times 10^{11} M_{\text{Sun}} \text{ (see note 2)}$$

[note 1] M_{vir} is defined as the mass enclosed in a sphere of mean density contrast of 360 with respect to the background density

[note 2] M_{200} is defined as the mass enclosed in a sphere of mean density contrast of 200 with respect to the critical density; this follows the definition in Power et al. 2003 so it can be used in the estimation in the next sections (see [this link](#) for more details)

3. How to Select Highest Resolution Cuboids in MUSIC ICs

The highest resolution cuboid (HRC) in the MUSIC initial condition is determined so that the target halo at $z=0$ is resolved with finest resolution particles. HRC is defined by `ref_offset` and `ref_extent` in the MUSIC parameter. Please note that the highest resolution volume in MUSIC can **only** be a cuboid.

3-A. Different Criteria to Select the HRC in MUSIC ICs

(1) HRC-1 = (`ref_offset`=0.622, 0.554, 0.438, `ref_extent`=0.032, 0.045, 0.018)

HRC-1 uses the Lagrangian cuboid of a sphere of radius $2 \times (\text{maximum extent of the target halo at } z=0) \sim 3 \times (\text{virial radius of the halo})$. It does **not** guarantee that the target halo at $z=0$ consists of only the finest resolution particles; i.e. target halo at $z=0$ could be contaminated by more massive particles. You may want to subtract the initial average velocity of the HRC by setting `center_vel=yes`.

(2) HRC-2 = (`ref_offset`=0.619, 0.554, 0.417, `ref_extent`=0.040, 0.058, 0.039, numbers not finalized yet)

While HRC-1 may work well in particle-based codes, it is often observed in grid-based codes that the target halo moves out of the region of the highest refinement. This problem tends to be more severe for low-mass halos which on average move faster than high-mass counterparts. Setting `center_vel=yes` alleviates the problem a bit, but not entirely. HRC-2 resolves this issue by adding extra layers to HRC-1. To check whether the additional buffer is thick enough to prevent the target halo from moving too close to the edges of the HRC and yet it is the thinnest possible, one has to rely on running the simulations multiple times. Still, it does **not** guarantee that the target halo at $z=0$ consists of only the finest resolution particles. Naturally, HRC-2 is larger than HRC-1. In fact, it is very close to HRC-3 in size, so the advantage of choosing it over HRC-3 is only very limited.

(3) HRC-3 = (`ref_offset`=0.618, 0.55, 0.408, `ref_extent`=0.048, 0.065, 0.047)

Depending on science questions being pursued, one may want to avoid the contamination of the target halo by massive particles at $z=0$. HRC-3 is the smallest such cuboid that prevents massive particles from contaminating the target halo at $z=0$. In other words, the target halo at $z=0$ entirely consists of the finest resolution particles. A binary search on `ref_extent` and `ref_offset` of all three axes of the HRC is needed, which in many cases translates into multiple test simulations at low resolution. Naturally, HRC-3 is larger than HRC-2.

3-B. Cautionary Notes and Test Results

(1) The behavior of the particles may vary by code, resolution, and runtime parameters. Because the tests for `ref_extent` and `ref_offset` were performed with Enzo at relatively low resolution, the proposed HRCs may not behave as expected in other codes or with different resolution.

(2) It is worth to note again that, as of this writing, the highest resolution volume in a MUSIC IC can only be a cuboid. The rectangular shape of the IC might be less of a problem in grid-based codes if the refinement is adaptive, and/or the finest refinement region reshapes itself as the Lagrangian volume of the halo shrinks. However, the shape of HRC would make the computation expensive, especially for a straightforward version of particle-based codes. Therefore, in order to expedite the calculation, one may consider masking out the particles that do not contribute to the target halo at $z=0$, based on their own low-resolution tests in each code.

(3) Volume of the HRC: HRC-1 (=18 % of HRC-3) < HRC-2 (=62 % of HRC-3) < HRC-3

(4) Halos found in Enzo test runs: (writing in progress)

(5) Levels of contamination measured in Enzo test runs: (writing in progress)

3-C. Support for Minimum Volume High-resolution Regions <-- NEW!! (written by Oliver Hahn, 03/22/2013)

The latest version of MUSIC supports minimum bounding ellipsoids enclosing a supplied point set of Lagrangian coordinates to define the high resolution region. Highest-resolution output for particle codes will be generated only inside the ellipsoid mask, output for RAMSES will include the mask as a separate mask file, ENZO is not yet supported.

As an example of how to use this feature, you can use this [region_point_file](#) of the $2R_{\text{vir}}$ Lagrangian volume obtained with a Gadget-2 simulation. The necessary options to generate ellipsoid ICs using this file are specified in section [setup]. The parameters `region_point_shift` and `region_point_levelmin` (below) are specific to the simulation from which the Lagrangian coordinates have been obtained:

```

region = ellipsoid
region_point_file = region_agora.txt # for the file downloadable above
region_point_shift = -18, -10, 8
region_point_levelmin = 7

```

This file and parameters have been used in the Gadget-2/3 and Gasoline runs shown below in Section 6.

4. Simulation

It would be ideal for the flagship paper if we could compare RUN-2 at $z=0$ with all participating codes.

4-A. List of Simulation Parameters

	RUN-1	RUN-2
levelmax in MUSIC parameter	12	13
Mass resolution in the finest level	$237267 M_{\text{sun}}/h = 3.38 \times 10^5 M_{\text{sun}}$	$29658 M_{\text{sun}}/h = 4.22 \times 10^4 M_{\text{sun}}$
ref_offset, ref_extent in MUSIC parameter	HRC-3 or masked HRC-3 (see Section 3)	HRC-3 or masked HRC-3 (see Section 3)
SPH gravitational softening length in the highest-resolution region	3.22 comoving kpc until $z=9$, 322 proper pc afterwards (note 1)	1.14 comoving kpc until $z=9$, 114 proper pc afterwards
AMR finest resolution cell size	326 comoving pc (note 2) i.e. maximum refinement level = 11	163 comoving pc i.e. maximum refinement level = 12
AMR refinement criterion	particle over-density of 4	particle over-density of 4
Snapshots stored (at the very least)	$z = 6, 3, 2, 1, 0.5, 0.0$	$z = 6, 3, 2, 1, 0.5, 0.0$
Cautionary notes		Note that HRC-3 requires ~64GB of RAM in the white noise generation (Mike Kuhlen)

[note 1] $0.5 \cdot 4 \cdot R_{200} / \sqrt{N_{200}}$, from Mike Kuhlen's suggestion of 50% of ϵ_{opt} found in Power et al. 2003
 $= 0.5 \cdot 4 \cdot (1.7e11 \cdot 2e33)^{1/3} \cdot [200 \cdot (4 \cdot 3.14/3) \cdot 1.88e-29 \cdot (0.702)^2]^{-1/3} / 3.086e18 / (1.7e11/3.38e5)^{1/2} = 322 \text{ pc}$

Therefore,

-> $\epsilon = 322 \text{ pc}$ for the highest-resolution region

-> $\epsilon = 322 \text{ pc} \cdot 8^{[(\text{level_max} - \text{level_here})/2]}$ for the lower resolution regions because ϵ goes as $\sim 1/\sqrt{N} \sim \sqrt{m_{\text{particle}}}$

[note 2] $60 \text{ Mpc}/h/2^7/2^{11} = 326 \text{ pc}$, where level = 0 is the root grid with 2^7 cells, i.e. levelmin = 7 in the MUSIC parameter

[note 3] Notice the units used. pc means pc, not pc/h; M_{sun} means M_{sun} , not M_{sun}/h

4-B. Proposed Simulation Strategy

- (1) First perform RUN-1 down to $z=0$ with HRC-3 (cuboid) or masked HRC-3 (ellipsoid)
- (2) Each group reports results and problems, and establishes ways to expedite the calculation
- (3) Move on to performing RUN-2 with HRC-3 or masked HRC-3
- (4) Compare the simulations at $z=0$ (or fallback to $z=1$?) and publish the result in the Flagship paper

5. Analysis

Below are tentative suggestions. cgs units are preferred to avoid any confusion across different codes. Once we agree on these items we may move on to refining the figures, or possibly to other diagnostics.

- (1) **Halo mass:** M_{vir} at $z=0$, where M_{vir} is defined as the mass enclosed in a sphere of mean density contrast of 360 with respect to the background density (see the definition in [Section 2](#))
- (2) **Halo center:** the location of the center of mass of the target halo identified as above, in the coordinate between [0.0, 0.0, 0.0] and [1.0, 1.0, 1.0]; halo center should approximately agree if the same HRC is used.
- (3) **Particle projection:** density-weighted projection of dark matter density along the z -axis at $z=0$ in a 1 Mpc/h box centered on the target halo, with only the finest resolution particles in a slab of 1 Mpc/h depth, colored by the density in the range of $[10^{-32}, 10^{-25}] \text{ g/cm}^3$
- (4) **Dark matter density profile:** radial profile of dark matter density at $z=0$ in spherical shells at 30 datapoints logarithmically spaced between 100 pc and 200 kpc, measured in the unit of g/cm^3 , do *not* exclude particles that belong to subhalos
- (5) **Subhalo mass (V_{max}) function:** (explicit descriptions/directions being built by Annalisa Pillepich et al.)
- (6) All other analysis including comparison of the halo catalogues, dark matter merger histories, power spectrum, etc. will go into the companion paper to introduce the IC in detail.

6. Results

All the numbers and figures are measured at $z=0$ unless marked otherwise. Please check the definitions/directions for all quantities in the previous section.

6.1. RUN-1 (~300 pc resolution)

(1) Summary^[1]

CODE	HRC Used	Resolution /Softening	Halo Mvir (Msun)	Halo Center	Particle Projection (z-axis)	Density Profile	Mass Function
ART-Chicago <i>Leitner</i>	masked 2 Rvir ^[3]	see table 4-A			6Mpc/h (to be made in yt shortly)		
ART-Chicago <i>Leitner</i>	HRC-3	see table 4-A					
ART-NMSU <i>Klypin</i>		see table 4-A					
ChaNGa <i>Quinn</i>		see table 4-A					
Enzo ^[2] <i>Kim</i>	HRC-3 vel.shift	see table 4-A	1.589e11	(0.492979, 0.507911, 0.507273) different coord. ^[4]	1Mpc/h 3Mpc/h 6Mpc/h (projected in a cube)	profile	
Gadget3-AFS <i>Hobbs</i> <i>/Iannuzzi</i>	masked 2 Rvir ^[3]	adaptive (without TreePM)		different coord.	1Mpc/h 3Mpc/h 6Mpc/h (projected in a cube) [1Mpc/h - yt3.0] ^[7]	profile	
Gadget3-CFS <i>Hobbs</i>		constant comoving 322 pc (without TreePM)					
Gadget3	HRC-3	see table	1.675e11	(0.486773,	1Mpc/h 3Mpc/h 6Mpc/h	profile	

<i>Nagamine /Thompson /Todoroki</i>		4-A (with TreePM) ^[5]		0.525076, 0.491379)	(projected in a cube) [1Mpc/h – yt3.0] ^[7]		
Gadget2-CFS <i>Hahn</i>	masked 2 Rvir ^[3]	constant comoving 322 pc (with TreePM)	1.660e11	(0.4959219, 0.5354019, 0.4714843) shifted coord.	1Mpc/h 3Mpc/h 6Mpc/h (projected in a cube) [1Mpc/h – yt3.0] ^[7]	profile	
Gasoline <i>Guedes</i>	masked 2 Rvir ^[3]	see table 4-A			1Mpc/h 3Mpc/h 6Mpc/h (projected in a cube)		
Gasoline <i>Shen /Pillepich</i>	HRC-3	see table 4-A	1.650e11	(0.485262, 0.526979, 0.489465) ^[6]	1Mpc/h 3Mpc/h 6Mpc/h (projected in a cube) [1Mpc/h – yt3.0] ^[8]		
Nyx <i>Schmidt</i>	HRC-3	see table 4-A					
Pkdgrav2 <i>Kuhlen</i>	HRC-3	see table 4-A		(0.509149, 0.524213, 0.457254) ^[6]	1Mpc/h 3Mpc/h 6Mpc/h (projected in a cube) [1Mpc/h – yt3.0] ^[9]		
Ramses <i>Hahn /Teyssier</i>	masked 2.5 Rvir ^[3]	see table 4-A	1.675e11	(0.486123, 0.526438, 0.490137)	1Mpc/h 3Mpc/h 6Mpc/h (projected in a cube) [1Mpc/h – yt3.0] ^[10]	profile	
Ramses <i>Hahn /Teyssier</i>	HRC-3	see table 4-A	1.665e11	(0.485985, 0.526657, 0.489847)	1Mpc/h 3Mpc/h 6Mpc/h (projected in a cube)	profile	

[note 1] Results here are very preliminary at this stage and subject to change a lot.

[note 2] Note for Enzo users: RefineRegionAutoAdjust=0, PotentialIterations=20, enzo-dev on bitbucket (changeset 99d895b29db1, Jan. 2013)

[note 3] See Section 3–C above for details of ellipsoidal ICs.

[note 4] The halo center is found naturally at a different coordinate, because this IC used a constant velocity shift across the entire domain.

[note 5] For more information, and variations of this run to test the TreePM switch in Gadget and softening schemes, see <http://www.physics.unlv.edu/~keitee/Agora.html> and <http://www.physics.unlv.edu/~keitee/AGORA/AGORA-RUN1-results.pdf>

[note 6] The native coordinate system for the Topsy format (Gasoline/Pkdgrav2) is [–0.5, 0.5].
The halo center here is manually corrected by [+0.5, +0.5, +0.5] to match the convention of others.

[note 7] Detailed yt–3.0 usage for Gadget [here](#) and the [yt script](#) used, thanks greatly to Matthew Turk

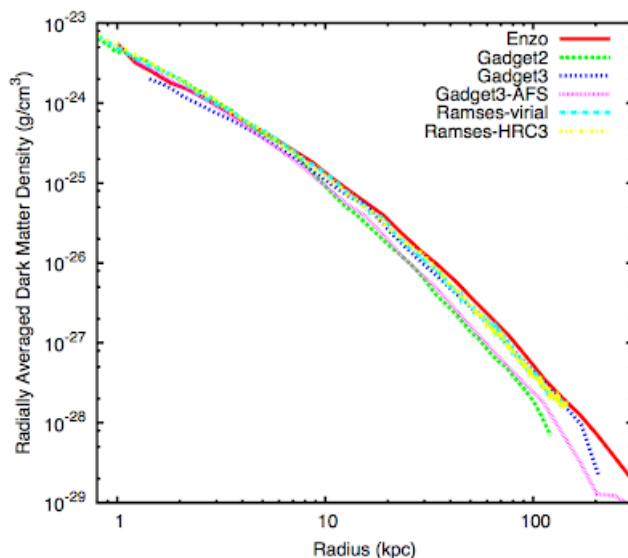
[note 8] Detailed yt–3.0 usage for Gasoline [here](#) and the [yt script](#) used, thanks greatly to Matthew Turk

[note 9] Detailed yt–3.0 usage for Pkdgrav2 [here](#) and the [yt script](#) used, thanks greatly to Matthew Turk

[note 10] Detailed yt–3.0 usage for Ramses [here](#) and the [yt script](#) used, thanks greatly to Matthew Turk

(2) Composite Dark Matter Density Profile

(preliminary study in progress by Annalisa Pillepich, Ji-hoon Kim, et al.)



(3) Composite Subhalo Mass Function

(preliminary study in progress by Annalisa Pillepich et al.)

6.2. RUN-2 (~100 pc resolution)

(1) Summary^[1]

CODE	HRC Used	Resolution /Softening	Halo Mvir (Msun)	Halo Center	Particle Projection (z-axis)	Density Profile	Mass Function
ART-Chicago <i>Leitner</i>		see table 4-A					
ART-NMSU <i>Klypin</i>		see table 4-A					
ChaNGa <i>Quinn</i>		see table 4-A					
Enzo ^[2] <i>Kim</i>	HRC-3	see table 4-A	1.596e11	(0.484711, 0.525988, 0.486506)	1Mpc/h 3Mpc/h 6Mpc/h (projected in a cube)	profile	
Gadget3-AFS <i>Hobbs</i> <i>/Iannuzzi</i>		adaptive (with TreePM)					
Gadget3-CFS <i>Hobbs</i>		constant comoving 114 pc (with TreePM)					
Gadget3 <i>Nagamine</i> <i>/Thompson</i>	HRC-3	see table 4-A (with	1.704e11	(0.484631, 0.525307, 0.489904)	1Mpc/h 3Mpc/h 6Mpc/h	profile	

<i>/Todoroki</i>		TreePM)					
Gadget2-CFS <i>Hahn</i>		constant comoving 114 pc (with TreePM)					
Gasoline <i>Guedes</i>		see table 4-A					
Gasoline <i>Shen</i> <i>/Pillepich</i>	HRC-3	see table 4-A	1.650e11	(0.482979, 0.527152, 0.487906)	1Mpc/h 3Mpc/h 6Mpc/h		
Nyx <i>Schmidt</i>		see table 4-A					
Pkdgrav2 <i>Kuhlen</i>	HRC-3	see table 4-A	1.718e11	(0.483358, 0.527631, 0.487607)	1Mpc/h 3Mpc/h 6Mpc/h	profile (plot)	
Ramses <i>Hahn</i> <i>/Teyssier</i>	masked 2.5 Rvir	see table 4-A	1.656e11	different coord.	1Mpc/h 3Mpc/h 6Mpc/h	profile	

[note 1] Results here are very preliminary at this stage and subject to change a lot.

[note 2] Note for Enzo users: RefineRegionAutoAdjust=0, PotentialIterations=20, enzo-dev on bitbucket (changeset 99d895b29db1, Jan. 2013)

[note 3] For more information for the Gadget run by the UNLV group, see

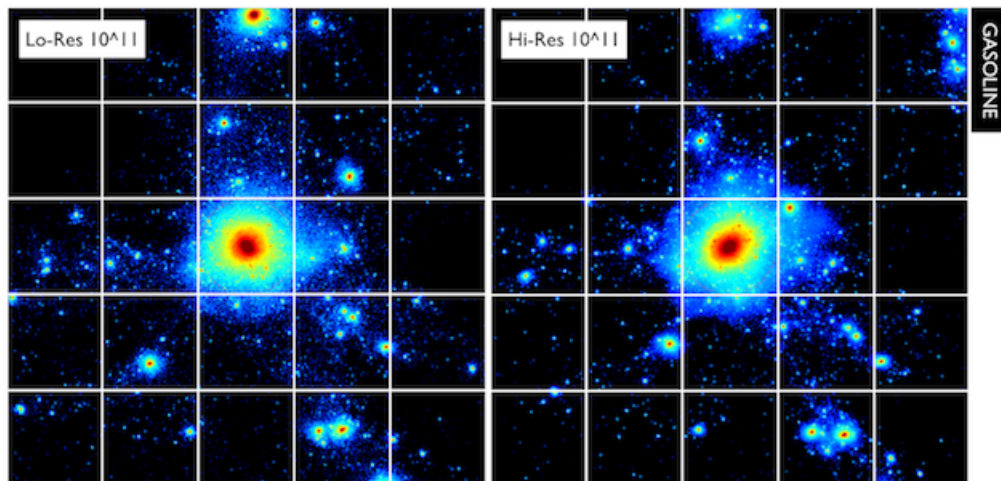
<http://www.physics.unlv.edu/~keitee/Agora.html> and <http://www.physics.unlv.edu/~keitee/AGORA/AGORA-RUN2-summary-table.pdf>

(2) Resolution Tests/Comparisons: GASOLINE runs with AmigaHaloFinder and Topsy

(preliminary study in progress by Annalisa Pillepich et al.)

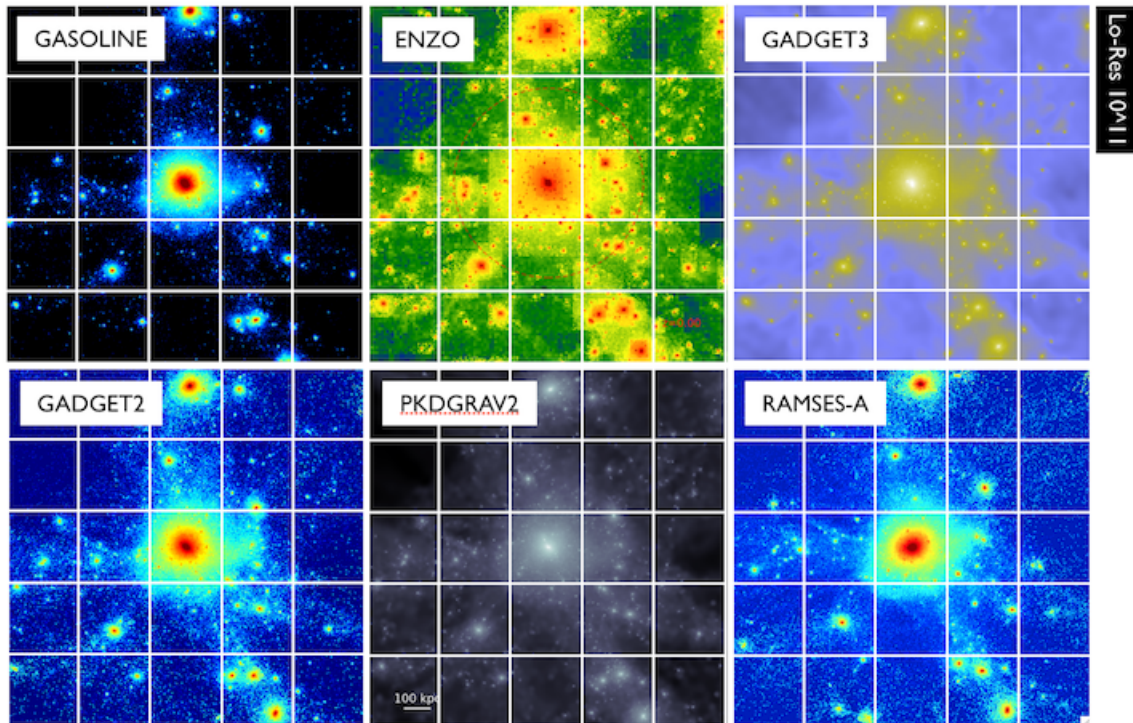
(2a) DM Density Projections

- The centers of the target halo are displaced by ~237 kpc between the two resolution runs
Q/TBD: How about other codes? What is this due to? How does this affect contamination?
- The resolution-induced discrepancies are visually manifest already at scales as big as ~1Mpc, for the same projections in cubes centered at the corresponding halo centers
(all this caveat properly matching volume-density color coding and smoothing kernels)



- It "seems" that inter-resolution discrepancies with Gasoline are more pronounced than (some) inter-code discrepancies at fixed

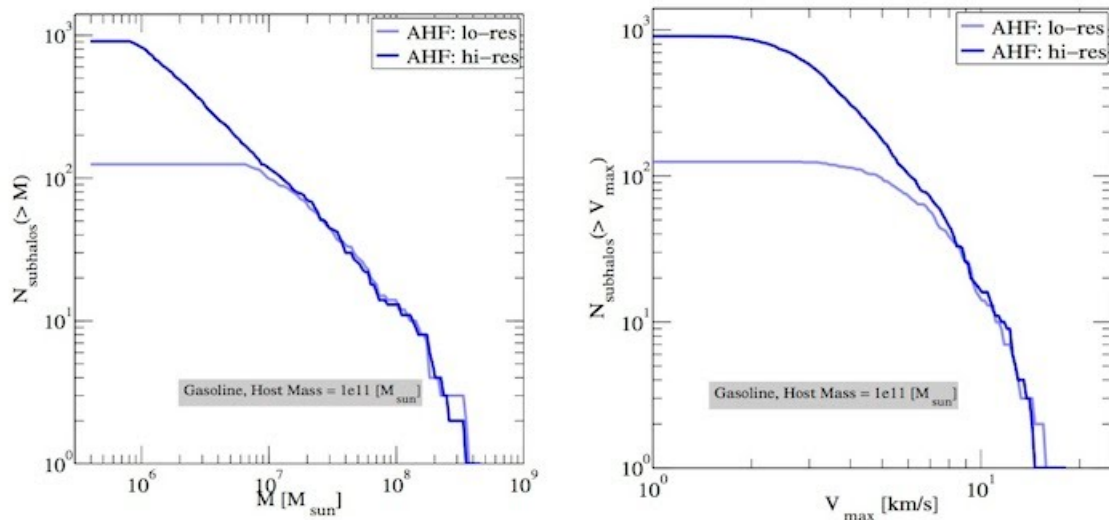
resolution; compare for example figure above to mosaic below:
(to be repeated with consistent density maps!)



(2b) DM Density Profiles

(2c) Subhalo Abundances

- Subhalo Abundance of the GASOLINE runs analyzed with the Amiga Halo Finder: convergence is decent.



- TBD: check V_{max} measurements and compare to other Halo Finders, i.e. Rockstar
- check [this page](#) for more information on halo finder adopted in this study!

(3) Composite Dark Matter Density Profile

(4) Composite Subhalo Mass Function

7. Participants

Here are the representatives from participating code groups. Many thanks to all the participants who stepped up to the plate for this proof-of-concept run. The default softening scheme is described in table 4-A.

	CODE	VOLUNTEER
1	ART-Chicago	Sam Leitner
2	ART-NMSU	Anatoly Klypin
3	CHANGA	Tom Quinn
4	ENZO	Ji-hoon Kim, John Wise
5	GADGET2-CFS (Constant Force Softening)	Oliver Hahn
6	GADGET3-CFS (Constant Force Softening)	Alex Hobbs, Justin Read
7	GADGET3-AFS (Adaptive Force softening)	Alex Hobbs, Justin Read, Francesca Iannuzzi
8	GADGET3 (softening switched at $z=9$)	Ken Nagamine, Robert Thompson, Keita Todoroki
9	GASOLINE	Sijing Shen, Javiera Guedes, Annalisa Pillepich
10	NYX	Jan Engels, Wolfram Schmidt
11	PKDGRAV2	Mike Kuhlen, Javiera Guedes
12	RAMSES	Oliver Hahn, Romain Teyssier